Text as Data
ifo Institute, 15–17 September 2026

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1. Overview

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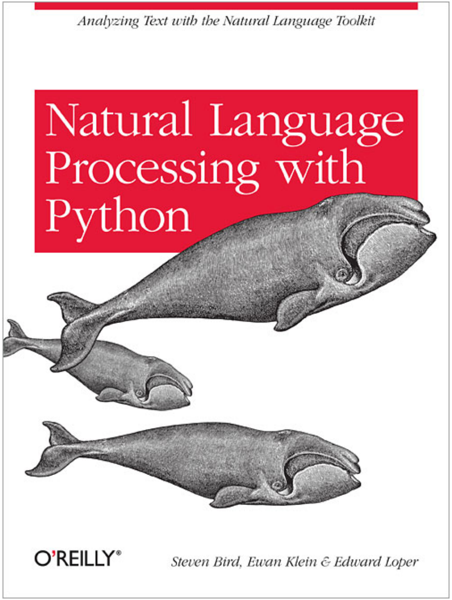
- Economics:
  - Relate text data to metadata to understand economic/political/social forces
  - e.g., analyze the motivations and decisions of public officials through their writings and speeches
  - Assess the real-world impacts of language on government and the economy

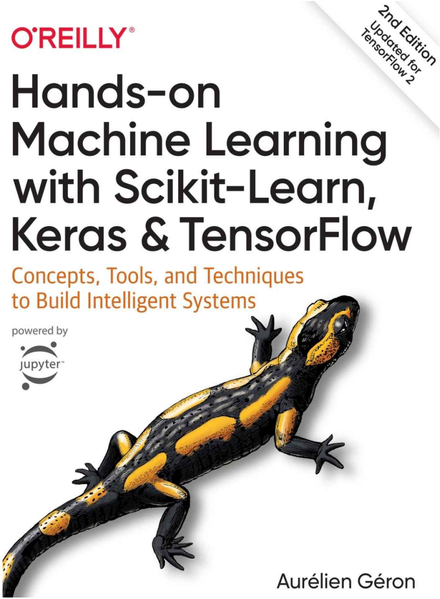
Syllabus and Materials

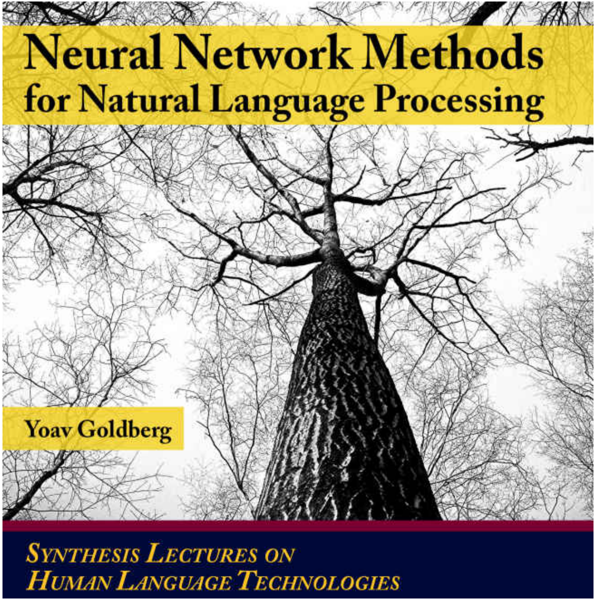
- Three half-days, 15–17 September 2026, 9:00–13:00
- Course syllabus (readings, schedule):
  - Syllabus – Text as Data in Economics, ifo 2026 (Google Doc)
- Course repository (slides, code, papers – updated before each session):
  - `github.com/BenjaminArold/Text-as-Data-ifo-2026`

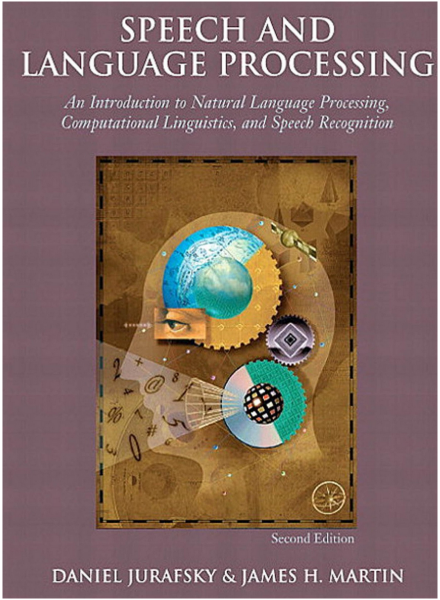
Course Bibliographies

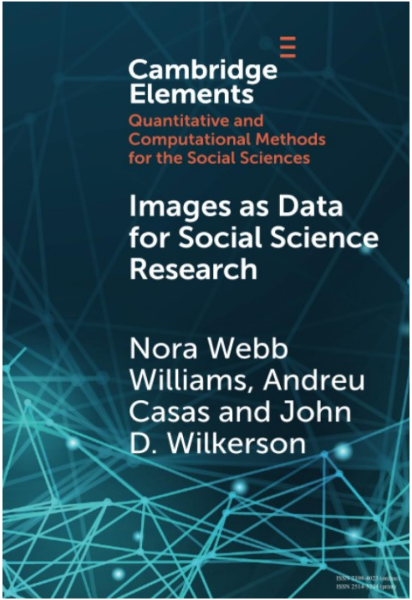
- Materials referenced in syllabus:
  - Lecture slides
  - Papers (core and suggested reading)
  - Additional Material


Analyzing Text with the Natural Language Toolkit
Natural Language Processing with Python
O'REILLY®
Steven Bird, Ewan Klein & Edward Loper


O'REILLY
2nd Edition
Updated for TensorFlow 2
Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow
Concepts, Tools, and Techniques to Build Intelligent Systems
powered by
jupyter
Aurélien Géron


Neural Network Methods
for Natural Language Processing
Yoav Goldberg
SYNTHESIS LECTURES ON
HUMAN LANGUAGE TECHNOLOGIES


SPEECH AND
LANGUAGE PROCESSING
An Introduction to Natural Language Processing,
Computational Linguistics, and Speech Recognition
Second Edition
DANIEL JURAFSKY & JAMES H. MARTIN


Cambridge
Elements
Quantitative and
Computational Methods
for the Social Sciences
Images as Data
for Social Science
Research
Nora Webb
Williams, Andreu
Casas and John
D. Wilkerson

Main Python packages for NLP

- Python 3 is ideal for text data / natural language processing, image analysis, and audio analysis
  - Can use Anaconda or download the packages we need to a pip environment
    - nltk – broad collection of pre-neural-nets NLP tools
    - scikit-learn – ML package with nice text vectorizers, clustering, and supervised learning
    - xgboost – gradient-boosted machines for supervised learning
    - gensim – topic models and embeddings
    - spaCy – tokenization, NER, parsing, pre-trained vectors
    - huggingface / sentence-transformers / BERTopic – pre-trained transformers, document embeddings, embedding-based topic models
    - tensorflow / keras – deep learning-based text/image/audio analysis
    - librosa - library for audio analysis
  - Code samples (based on Ash & Hansen, 2023: Text algorithms in economics. Annual Review of Economics, 15, 659-688.:): https://github.com/BenjaminArold/Text-as-Data-ifo-2026/tree/main/code)

Logistics

Learning Materials

Course Content Overview

Text is high-dimensional

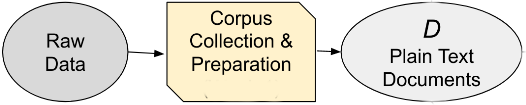
- Sample of documents, each n_L words long, drawn from vocabulary of n_V words
- The unique representation of each document has dimension $n_V^{n_L}$
  - e.g., a sample of 30-word Twitter messages using only the one thousand most common words in the English language
    - $\rightarrow$ dimensionality $= 1000^{30} = 10^{32}$

“Text as Data”, GKT 2017

Summarize analysis in three steps:

- convert raw text D to numerical array $\boldsymbol{C}$
  - The elements of $\boldsymbol{C}$ are counts over tokens (words or phrases)
- map $\boldsymbol{C}$ to predicted values $\hat{\boldsymbol{V}}$ of unknown outcomes $\boldsymbol{V}$
  - Learn $\hat{\boldsymbol{V}}(\boldsymbol{C})$ using machine learning
  - e.g. supervised learning for some labeled $\boldsymbol{C}_i$ and V_i
  - or unsupervised learning of topics/dimensions just from $\boldsymbol{C}$
- use $\hat{\boldsymbol{V}}$ for subsequent descriptive or causal analysis

Corpora



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- The set of documents is the corpus, which we will call D.

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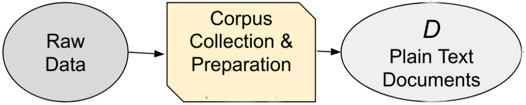

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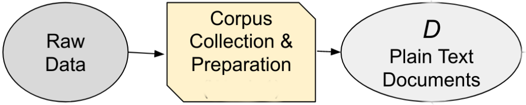

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- Text data is **unstructured**:
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- All text data approaches will throw away some information:
  - The trick is figuring out how to retain valuable information

This course is about relating documents to metadata

- This course is on **analysis of text, image, and audio** data for **economics**:
  - start with text analysis (natural language processing NLP), then transfer methods to image and audio
  - text documents are not that meaningful by themselves
  - we want to relate **text** data to **metadata**

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- e.g., measuring positive-negative sentiment Y in political speeches
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  - text documents are not that meaningful by themselves
  - we want to relate **text** data to **metadata**
- e.g., measuring positive-negative sentiment Y in political speeches
  - not that meaningful by itself
- but how about sentiment Y_{ijkt} in speech i by politician j on topic k at time t:
  - how does sentiment vary over time t?
  - does politician from party p_j express more negative sentiment toward topic k?

What counts as a document?

The unit of analysis (the "document") will vary depending on your question

- needs to be fine enough to fit the relevant metadata variation
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E.g., what should we use as the document in these contexts?

1. predicting whether a judge is right-wing or left-wing in partisan ideology, from their written opinions
2. predicting whether parliamentary speeches become more emotive in the run-up to an election

Topics Covered

Text-as-Data:

- Dictionaries, Tokenization, and Document Distance
- Topic Models and ML with text, Word Embeddings and Linguistic Parsing
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Ethical Considerations

Strengths and weaknesses of NLP methods

Method	Strengths	Weaknesses
NLP 0.0		
Reading	Captures meaning, nuance, context; uses domain expertise; transparent	Subjective; does not scale; non-replicable
NLP 1.0		
Dictionaries	Scalable, replicable, transparent; cheap; extensible to historical text	Misses context (e.g., negation); requires good dictionary
NLP 2.0		
Embeddings	Captures implicit relations; substrate of all modern LLMs	Black box; requires training data; pre-trained corpus may not fit domain
NLP 2.5		
Grammar parsing	Captures syntactic structure; precise on negation/modality; legally meaningful	Requires good parser; brittle to OCR/text errors
NLP 3.0		
LLMs	Nuanced; flexible; near-human judgment at scale	Expensive; opaque; version drift across vendors
NLP 4.0		
GenAI agents	Orchestrate multi-step pipelines; massive scale; combine all prior tools	Requires careful validation; selection in what's online; cost

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- Precision $\leftrightarrow$ Scale
  - Expert reading: high precision, low scale
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  - Dictionaries: fully replicable, less nuanced
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  - LLMs/GenAI agents: more nuanced, version-dependent, non-deterministic
- Practical takeaway: the methods are **complements, not substitutes**
  - the methods in this course validate and cross-check each other